COVID Blood Processsing for scRNAseq v4 01JUL20

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## 1. Objective

The isolation of lymphocytes and pan-mononuclear cells from human whole blood for scRNAseq analysis.

## 2. Materials

50mL Centrifuge Tube (Fisher Scientific, Cat. No.: 12-565-271)

5mL Falcon™ Round-Bottom Polypropylene Tubes (Fisher Scientific, Cat. No.: 14-959-11A)

Biotin anti-human CD235ab Antibody (100µg) (Biolegend, Cat. No.: 306618)

Biotin anti-human CD66b Antibody (100µg) (Biolegend, Cat. No.: 305120)

BioMag®Plus Streptavidin (Bangs Laboratories, Cat. No.: BP628)

Cryogenic Vials (Fisher Scientific, Cat. No.: 09-761-71)

Cryostor CS10 (Fisher Scientific, Cat. No.: NC9930384)

Dulbecco's Phosphate Buffered Saline (DPBS) (Fisher Scientific, Cat. No.: 14-190-144)

EDTA 0.5M pH 8.0 (Fisher Scientific, Cat. No.: 15-575-020)

Fetal Bovine Serum (FBS) (Fisher Scientific, Cat. No.: 10-099-14)

Ficoll-Paque™ PLUS Media (Fisher Scientific, Cat. No.: 45-001-749)

Human TruStain FcX (Biolegend, Cat. No.: 422302)

NC-Slide A8 (Chemometec, Cat. No.: 942-0003)

Penicillin-Streptomycin-Glutamine (100X) (Fisher Scientific, Cat. No.: 10-378-016)

Solution 13 AO/DAPI (Chemometec, Cat. No.: 910-3013)

Dead Cell Removal Kit (Miltenyi, Cat. No.: 130-090-101)

- Dead Cell Removal Microbeads
- 20x Binding Buffer Stock Solution

MS Columns (Miltenyi, Cat. No.: 130-042-101)

RosetteSep Granulocyte Depletion Cocktail (Stemcell Techonologies, Cat. No.: 15624)

## 3. Equipment

Centrifuge

Cell Counter - NC-3000

EasyEights™ EasySep™ Magnet (Stemcell Technologies, Cat. No.: 18103)

MACS Multistand (Miltenyi, Cat. No.: 130-108-934)

## 4. Protocol

### 4.1.1. Preparing Buffer

4.1.2. Create the following **DPBS Solution-EDTA** in a bottle of DPBS by using the table below:

| *Component* | *Volume (mL)* | *Starting Conc.* | *Final Conc.* |
|---|---|---|---|
| DPBS | 474 | - | - |
| FBS | 25 | 100% | 5% |
| EDTA | 1 | 0.5M | 1mM |

*Table 1.*

4.2. **Biosafety Notes**

- All materials required for sample processing are to be prepared and in the biosafety cabinet before handling of blood samples.
- All blood sample manipulation takes place in a biosafety cabinet unless specificially stated.
- All centrifugation steps must take place in capped containers. Upon completion of a centrifugation step, return the capped container to the biosafety cabinet remove all tubes, and spray and wipe them with >70% ethanol before continuing to the next step.

4.3. **Preparation of Blood**

4.3.1. Place sample box into biosafety cabinet.

4.3.2. Remove samples from box and from the containment bags, discard bags, spray sample containers with >70% ethanol and wipe down.

4.3.3. Record the total volume of whole blood to be processed below:

__________mL

4.3.4. Spin the whole blood 400 x g for 10 minutes, 20°C in the anti-coagulant tubes and remove the plasma layer to cryovials. Be sure to not remove any red blood cells with the plasma. **Add the same volume to all cryovials.**

Record number of vials: __________ and the volume per vial __________ mL.

**NOTE:** Ensure that cryovials are decomtaminated prior to removal from the biosafety cabinet.

4.3.5. Add the same volume of **DPBS Solution-EDTA** as removed plasma back to the blood tube. Pipette to mix.

4.4. **RosetteSep and Ficoll-Paque**

4.4.1. Add 50µL of RosetteSep Cocktail/1mL of blood directly to the blood tube. Pipette to mix

4.4.2. Inclubate samples at room temperature for 20 minutes.

4.4.3. Transfer sample to a 50mL tube and dilute up to 25ml with **DPBS Solution-EDTA solution.**

4.4.4. Aliquot 15mL of Ficoll-Paque Media PLUS to a 50mL tube.

4.4.5. Using the **slow** setting on the pipette gun, **gently** layer the blood/DPBS Solution-EDTA mixture on top of the 15mL of Ficoll-Paque Media PLUS. **Take extra care not to disturb the blood-ficoll**

**interface while layering. Disturbing the interface excessively prevents the mononuclear cells from becoming a clean layer.**

4.4.6. Spin for 20 minutes, 1200 x g at 20°C with no brake, 4 acceleration.

**Note:** Centrifuge should be pre-warmed to 20°C.

4.4.7. Remove the mononuclear cell layer (PBMC) from each tube and transfer to a new 50mL tube. **Take extra to care to avoid pulling cells from the ficoll layer (underneath the mononuclear cell layer) as this typically contains a lot of granulocytes.** Pulling from the plasma layer is not an issue.

4.4.8. Top PBMC with cold DPBS Solution-EDTA to 40mL (ensure at least 2-3 volumes are added) and centrifuge the cell suspension(s) for 10 minutes at 400 × g, 4°C.

4.4.9. (Platelet Spin) Discard the supernatant, top tube to 40mL with cold DPBS Solution-EDTA, and centrifuge the cell suspension for 10 minutes at 120 × g, 4°C.

4.4.10. Remove the supernatant (**caution: pellet may be loose**), and re-suspend the cell pellet in 4mL **Dulbecco's Phosphate Buffered Saline (DPBS)**.

## 4.5. Cell Counting of COVID Samples

4.5.1. Add 0.050mL of sample, 0.050mL of DPBS, and 0.005mL of Solution 13 to a 1.5mL centrifuge tube, incubate for 2 minutes at room temperature.

4.5.2. Add 0.100mL of BD Cytofix Fixation Buffer to the samples and incubate 30 minutes, 4°C, and protect from light.

4.5.3. Aliquot 0.010mL of sample to the well of a NC-Slide A8 and count on the NC-3000.

Record number and viability below, calculate total cells:

cell number: __________cells/mL,___________% viable

## 4.6. Division of Sample for Analysis and Freeze-down

4.6.1. Aliquot up to 1 x $10^7$ cells to a 5mL Falcon Round-Bottom tube and place on ice for subsequent sample clean-up (Step 4.9).

4.6.2. Freeze down up to 1 x $10^8$ cells in approximately 1 x $10^7$ aliquots (1mL each) using Cryostor CS10 Medium, a Mr. Frosty, and a -80°C freezer. Record the number of vials frozen: __________ and the cells per cryovial frozen: _________.

4.7. Sample Clean Up for scRNAseq – CD66b and CD235ab removal

4.7.1. Centrifuge the single cell suspension for 5 minutes at 400 × g, 4°C.

4.7.2. Discard the supernatant and resuspend the cell pellet in 50L of DBPS Solution-EDTA; add 10µL of Human TruStain FcX to the single cell suspension and incubate for 10 minutes, 4°C.

4.7.3. Add 10µL of biotinylated anti-CD66b and biotinylated anti-CD235ab to the sample and incubate for 30 minutes, 4°C.

4.7.4. While the single cell suspension is incubating add 0.200mL of BioMag Plus Streptavidin Beads to a 5mL Falcon Round-Bottom tube.

4.7.5. Add 2mL of DPBS Solution-EDTA to the BioMag Plus Streptavidin Beads and place on a magnet for 5 minutes.

4.7.6. Remove all the supernatant from the BioMag Plus Streptavidin Beads, remove from the magnet and resuspend the beads in 0.100mL of DPBS Solution-EDTA.

4.7.7. Once step 4.73 is complete, add 3mL of DPBS Solution-EDTA to the single cell suspension and centrifuge for 5 minutes at 400 × g, 4°C.

4.7.8. Resuspend the single cell suspension in the BioMag Plus Streptavidin Beads from step 4.7.6, and incubate at room temperature for 5 minutes.

4.7.9. Add 3mL of DPBS to the tube and place on a magnet for 5 minutes.

4.7.10. Remove supernatant from tube and transfer to a separate 5mL Falcon Round Bottom tube.

4.8. Sample Clean Up for scRNAseq – Dead Cell Removal

4.8.1. Centrifuge the single cell suspension for 5 minutes at 400 × g, 4°C, and discard supernatant.

4.8.2. Resuspend cell pellet in 0.100mL of Dead Cell Removal Microbeads, mix well, incubate, room temperature, 15 minutes.

4.8.3. While the cell suspension is incubating, place an MS Column onto the MACS Multistand and rinse with 0.500mL 1x Binding Buffer Solution.

4.8.4. Post incubation, apply cell suspension to the MS Column and capture the flow through in a 5mL Falcon Round Bottom tube. Rinse with 1.5mL of 1x Binding Buffer and capture in the same tube.

4.8.5. Centrifuge the single cell suspension for 5 minutes at 400 × g, 4°C, and discard supernatant.

4.8.6. Resuspend cell pellet in 0.500mL DPBS, and count cells.

4.9. Cell Counting of COVID Samples (10x)

4.9.1. Add 0.050mL of sample, 0.050mL of DPBS, and 0.005mL of Solution 13 to a 1.5mL centrifuge tube, incubate for 2 minutes at room temperature.

4.9.2. Add 0.100mL of BD Cytofix Fixation Buffer to the samples and incubate 30 minutes, room temperature, and protect from light.

4.9.3. Aliquot 0.010mL of sample to the well of a NC-Slide A8 and count on the NC-3000.

4.9.4. Record number and viability below, calculate total cells:

cell number: __________cells/mL,___________% viable

4.10. 10X Encapsulation

4.10.1. Follow the appropriate 10X protocol (Chromium Next GEM Single Cell 3' Reagent Kits v3.1 User Guide – Rev D) for encapsulation of cells from the airway sample.